CHUKWUEMEKA OSMUND AZODO

MACHINERY ENGINEER | ROTATING EQUIPMENT, CONTROLS & RELIABILITY | OIL & GAS

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B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ

24 September 2026

Recruitment Team
ExxonMobil Nigeria
Esso Exploration and Production Nigeria Limited, Lagos

Application: Experienced Machinery Engineer, Nigeria Deepwater Assets (Job Code 109016)

Dear Hiring Team,

I am applying for the Experienced Machinery Engineer position supporting the Nigeria Deepwater assets and FPSOs. I am a mechanical engineer with over ten years of oil and gas experience, all of it in machinery and rotating equipment engineering, currently the Machinery Engineer for the Oso Gas Hub, the largest offshore gas processing facility in the Seplat joint venture and part of the former Mobil Producing Nigeria asset base. I hold a B.Eng in Mechanical Engineering and my NYSC discharge certificate, and comfortably exceed the five years of oil and gas and machinery-focused experience the role calls for. My daily work runs on ExxonMobil-legacy standards and machinery, so I would join already fluent in how your assets are engineered and governed.

Understanding equipment condition and identifying reliability risk is the substance of my current role. I review equipment performance daily: vibration spectra and trends (rotordynamic, torsional and pulsation, supported by my Category I vibration analyst training), lube-oil analysis, compressor performance maps, turbine parameters and pump curves across HP and LP centrifugal compressors, turbo-expanders, gas turbines, cryogenic and API 610 pumps, working through SolarInsight, Seeq and PI XHQ, and I steward the actions that keep those machines at target reliability and availability.

I act as technical lead for machinery troubleshooting and recovery, including root cause analysis for complex failures. When a bundle change-out left a 130 MMSCFD compressor with severe vibration, I led the RCA with the OEM and in-house teams, mapped the contamination paths and ran sectioned borescope inspection of the cooler bundles; the cause proved to be debris-induced rotor unbalance, and piping cleaning, lube-oil flushing and re-inspection returned the machine to service within ISO vibration limits. A separate fleet-wide RCFA (fifteen years of records, using pooled failure-rate and Crow-AMSAA reliability-growth analysis) let me show the affected compressors failing roughly 17× more often than a comparable benchmark unit on the same platform, and trace the cause to a wrong-specification filter element fitted into a housing built for now-discontinued OEM hardware. I am now driving the closeout: OEM and alternative-supplier engagement on a validated fix, packaged into an eleven-action plan with named owners and dates.

Alongside troubleshooting, I review equipment and system changes, specifications and deviations; contribute to risk screening, HAZID and detailed risk assessments; witness machinery Factory Acceptance Tests; and support engineering surveillance, commissioning and start-up. I strengthened Oso's Pre-Startup Safety Review process to capture off-skid activities it had missed, now tailored to each bundle change-out. Safety is a consistent strength: I maintain an appreciable record of identifying and following up on hazards under Seplat's Hazard Identification & Control Scheme, and earlier received FIPL's Safety Sentinel and Safety Advocate of the Year awards. Digitalization is a strong, hands-on competency of mine from earlier roles: IIoT sensor-to-cloud monitoring (Modbus, OPC-UA), applied machine learning for failure prediction, and machinery controls and instrumentation, most notably an IIoT condition monitoring programme that cut operating cost by around ten per cent, a capability not yet applied at Oso.

I coordinate directly with OEMs, contractors and cross-functional teams on maintenance and repair programs, document completed work so others can build on it, and report equipment issues, exposures and improvement opportunities clearly and promptly to asset leadership.

My offshore experience to date is on fixed platforms rather than FPSOs. The machinery scope transfers directly: gas turbine driven compression, turbo-expanders, API 610 and cryogenic pumps, lube-oil and seal systems, and Honeywell CCC anti-surge and turbomachinery control, and I would come up the curve on the floating production context quickly. I am an ASQ Certified Reliability Engineer, carry a valid BOSIET with CA-EBS and an Offshore Safety Permit, am based in Nigeria, and available at one month's notice.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience can support the reliability and availability of machinery systems across the Nigeria Deepwater FPSOs.

Yours sincerely,

Chukwuemeka Osmund Azodo